

6. Compute the efficiency e of the power screw:

$$e = \frac{Fp}{2\pi T_u}$$

where F is the load applied to the screw in lbs,

p is the pitch in inches,

T_u is the raising torque in lb · in

7. Calculate the maximum speed you can raise a load given an input power in hp.

- (a) Find the rotational speed of the screw (in rpm) using:

$$n = \frac{63000P}{T}$$

where P is the input power in hp,

T is the raising torque in lb · in

- (b) Convert to linear speed (in in/s) using the following relation:

$$V = \left(\frac{n \text{ rev}}{\text{min}} \right) \left(\frac{p \text{ in}}{\text{rev}} \right) \left(\frac{\text{min}}{60 \text{ sec}} \right)$$

where n is the rotational speed in rpm,

p is the pitch in in

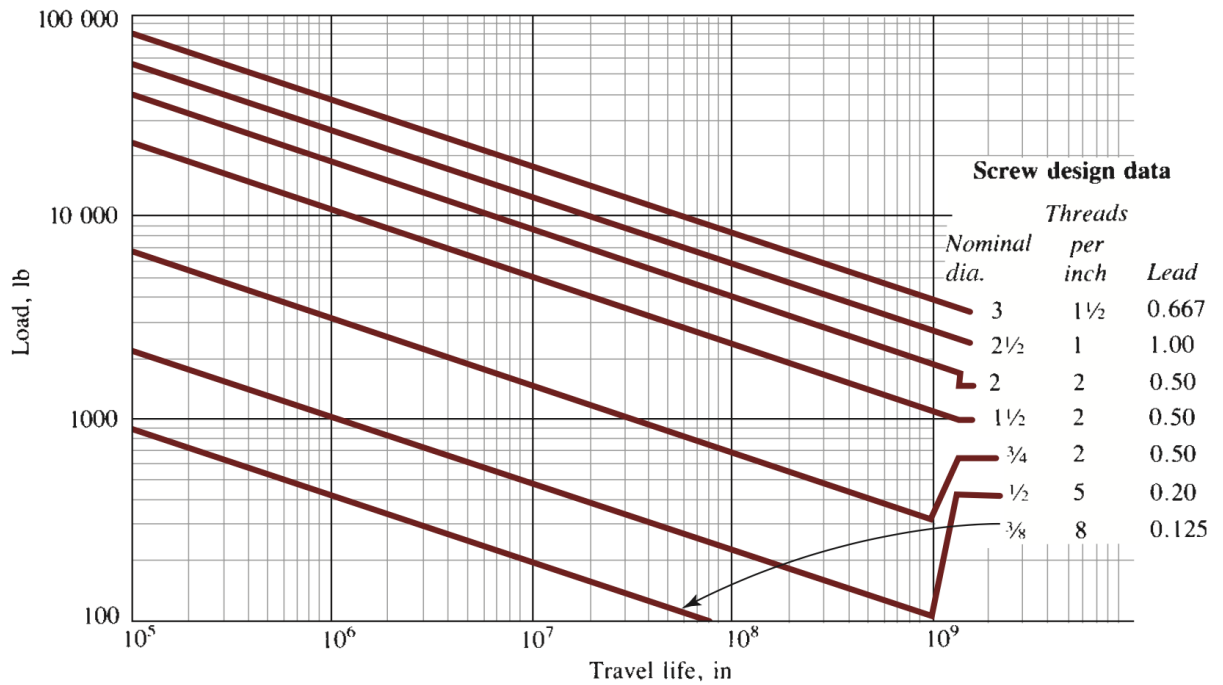
5.2 Ball Screws

5.2.1 Design Selection

1. Compute the travel life in inches:

$$\text{Travel Life} = \left(\frac{\text{distance (in)}}{\text{cycle}} \right) \left(\frac{\# \text{ of cycles}}{\text{hour}} \right) \left(\frac{24 \text{ hours}}{\text{day}} \right) \left(\frac{365 \text{ days}}{\text{year}} \right) (\# \text{ of years})$$

2. Select a ball screw based on load and travel life:



3. Compute the torque (in lb · in) required to drive the screw using the following equation:

$$T_u = 0.177FL$$

where F is the applied load in lbs,

L is the lead (from the graph above) in inches

4. Compute the power required (in hp) given a travel speed:

- (a) Calculate the rotational speed required (in rpm):

$$n = (V) \left(\frac{1 \text{ rev}}{L} \right) \left(\frac{60 \text{ sec}}{\text{min}} \right)$$

where V is the travel speed in in/s,

L is the lead in inches

- (b) The power required (in hp) is the following:

$$P = \frac{Tn}{63000}$$

where T is the required torque in lb · in,

n is the rotational speed in rpm

5.3 Springs

This section is best for Static Helical Springs, Chapter 10 of Shigley has pretty good depth on all spring types if we get something else